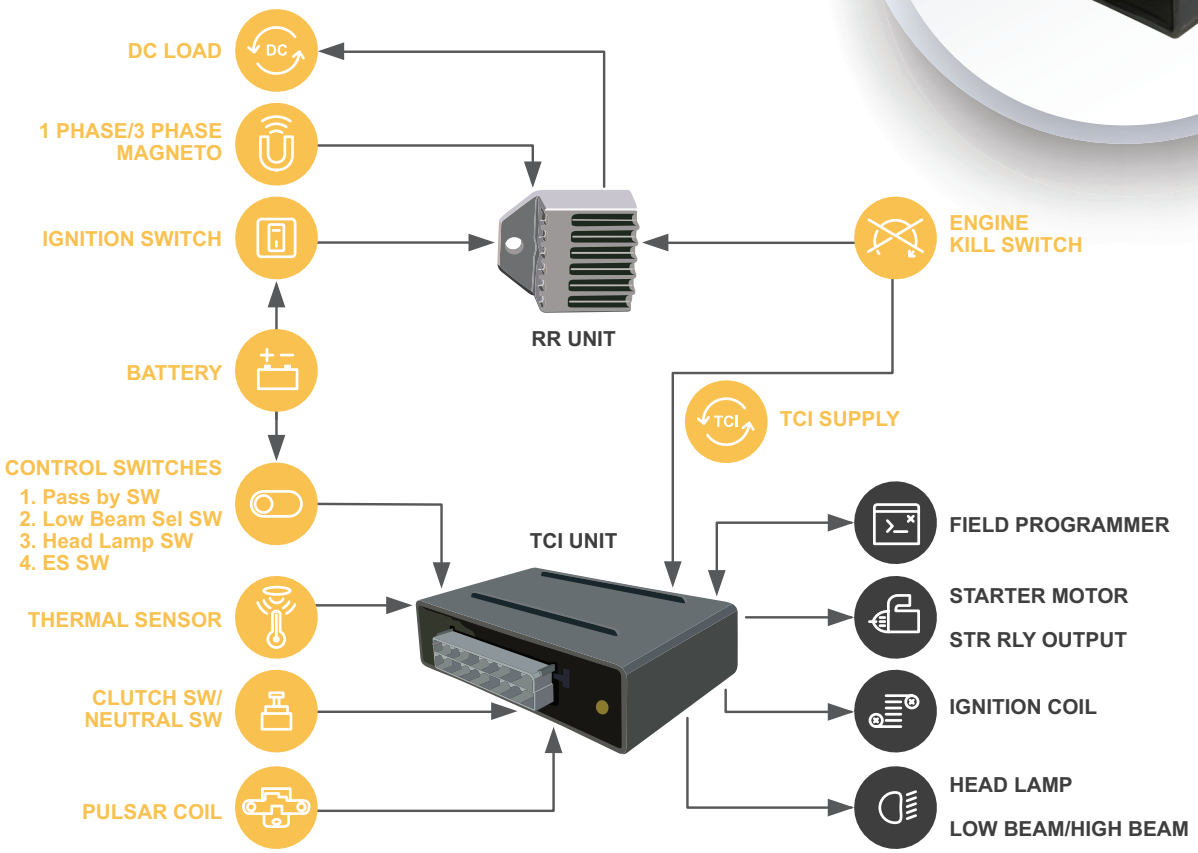
SEDEMAC™

Smart Ignition

SEDEMAC's Smart Ignition enables precise execution of ignition timing in 2 and 3 wheelers through real time dynamic estimation of load *without the use of a throttle position sensor.* This ensures optimal engine performance under varying speed and load conditions, using only crank position sensing. The technology leads to cost saving through elimination of throttle position sensor and associated wiring harness elements, while providing enhanced flexibility to OEMs in deciding ignition maps to better manage fuel economy, emissions and performance. This technology is applicable to all small spark ignited engines.

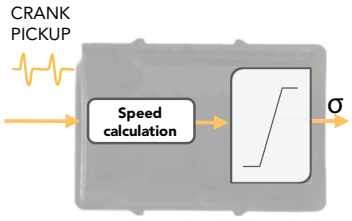
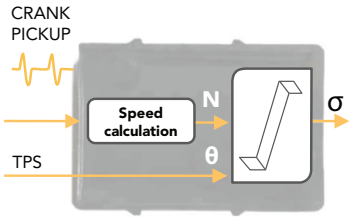
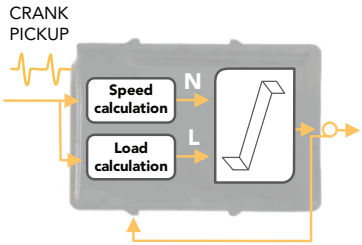
Smart Ignition is available as a standalone licensed software. It also comes in a package as ignition hardware (TCI/CDI) with licensed software. The software can be implemented on most digital ignition hardware that use microcontrollers.



SIC (Spark Ignition Controller)

SEDEMAC's Spark Ignition Controllers are of two types: TCI (Transistorised Controlled Ignition) and CDI (Capacitor Discharge Ignition). It has an electrical source, an ignition unit to store the source energy and to discharge the energy, sensors to indicate the instance of discharge time, and device to develop the required high voltage to emit spark in spark plug. **TCI/CDI** can be used for 2 wheelers, 3 wheelers, stationary gasoline engines, and lawn mowers etc. The controller generates the required high voltage spark and discharges it to the spark plug in the internal combustion engine, with precise ignition timing with high ignition capabilities. The controller is fully customizable according to each customers' requirements, and integrated with a starter relay control with respect to the rpm of the engine.

COMPARISON OF IGNITION STRATEGIES

ANALOG TCI/CDI	DIGITAL TCI/CDI + TPS	DIGITAL TCI/CDI + SEDEMAC SMART IGNITION
		
Age old technology used in India till 1999	*Used in India till 2012; Few OEMs still use it*	*Used by OEMs in India since 2012*

θ = Proxy for engine torque, L = Load estimation
N = Speed, TPS = Throttle Position Sensor